

LABID AL NAHIYAN

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EDUCATION

Bangladesh University of Engineering and Technology (BUET)
BSc in Computer Science and Engineering

Feb 2020 – Mar 2025

EXPERIENCE

TigerIT Bangladesh Ltd.

Machine Learning Engineer

August 2025 – Present

Dhaka, Bangladesh

- Reduced False Non-Match Rate (FNMR) by **41%** (from 0.0151 to 0.0089 @ FMR $\leq 10^{-2}$) for the **NIST MINEX III** benchmark by developing an optimized minutiae-based template generator.
- Engineered a **custom loss function** to address non-linear elastic distortions and rotational variances, significantly improving fingerprint matching accuracy and robustness in R&D production models.
- Architected a high-throughput data pipeline to preprocess and normalize **100,000+** biometric images, implementing automated feature extraction workflows to streamline large-scale training for full-fingerprint matching models.

RESEARCH & PUBLICATIONS

When Safety Blocks Sense: Measuring Semantic Confusion in LLM Refusals

Dec 2025

Co-Author

[arXiv:2512.01037](https://arxiv.org/abs/2512.01037)

- Introduced **"Semantic Confusion,"** a failure mode capturing local inconsistencies in LLM refusals where models reject harmless paraphrases of accepted intents.
- Developed **ParaGuard**, a 10k prompt corpus of controlled paraphrase clusters, and proposed token-level metrics (Confusion Index, Rate, and Depth) using embeddings and perplexity signals to diagnose safety-alignment gaps.

Contextually Appropriate Loss Function for Dense Crowd Tracking

Feb 2024 – Mar 2025

Undergraduate Thesis | Supervisor: Prof. A. B. M. Alim Al Islam, CSE, BUET

- Proposed a size-aware, gradient-stabilized loss function to optimize tracking of small, overlapping objects in dense crowds.
- Integrated the function into **YOLO-based models** and evaluated on JHU-CROWD++ and SCUT-HEAD, achieving superior mAP@50 and F1-scores compared to CIoU, DIoU, and GIoU.

SKILLS

Programming Languages

Python, C++, SQL, Java, TypeScript

ML & Frameworks

PyTorch, Hugging Face, OpenCV, NumPy, PostgreSQL, React.js, Express.js

Tools & Infrastructure

Docker, Linux (Ubuntu), GitHub, Postman

PROJECTS

Shop Genie - AI-Driven E-commerce Platform [SvelteKit, LLM, RAG]

[Github](#)

- Developed an AI-powered shopping assistant enabling users to find products through natural language search
- Scraped data for over 10,000 products across various shopping websites, enabling fast and robust product search

NerdHerd - All in one Educational Platform [SvelteKit, TailwindCSS, PostgreSQL]

[Github](#)

- Built an all-in-one student platform with classes with video calls, community features, quizzes, etc. Handled frontend and integrated frontend-backend systems seamlessly

Subset of C compiler [C, Bison, YACC, Flex]

[Github](#)

- Designed and implemented core components of a C compiler, including lexical analysis, parsing, symbol table management, and code generation using Flex, Bison, and YACC

ACHIEVEMENTS

- Champion - OpenAPI Hackathon, 11th National ICT Fest 2024 (IUT)
- 7th Place - SEC Inter-University Junior Programming Contest 2022 (Team: BUET Fast & Furious)